

# Apollo

Short audio clip  $\longrightarrow$  song title + source time

Noisy recordings, speed edits and covers

**Presenter A & Presenter B**

CSE 220 final project | September 2026

# How a clip becomes an answer

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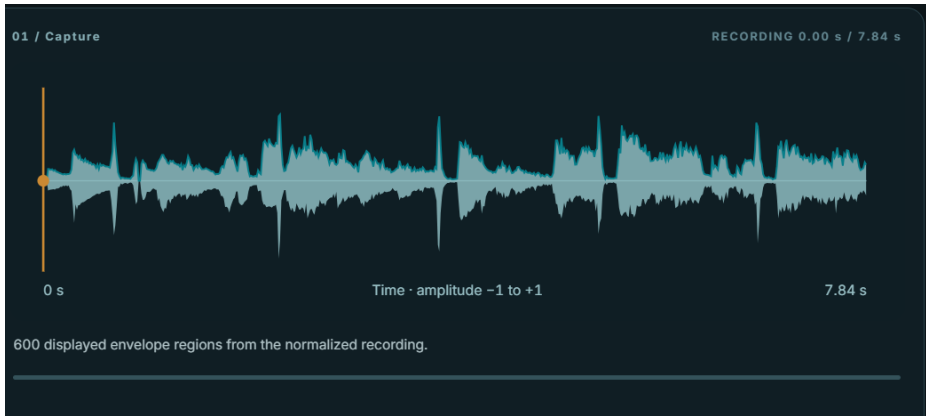
**Answer back to the page: song title + where the clip begins**

Signal code builds fingerprints; SQLite searches an indexed catalog.

# From sound to spectrogram

## 1 / 2 Capture the waveform

time →; height = signal amplitude

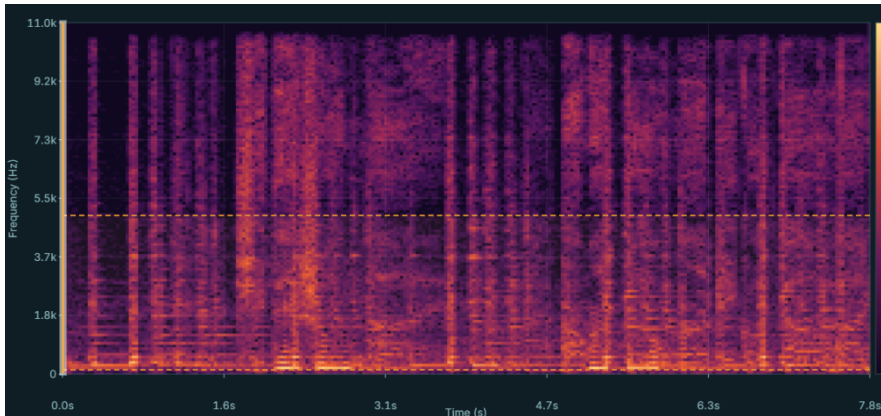


Mono + peak normalization + 22,050 samples/second for both song and query.

# From sound to spectrogram

## 2 / 2 STFT makes a frequency map

time  $\rightarrow$ , frequency  $\uparrow$

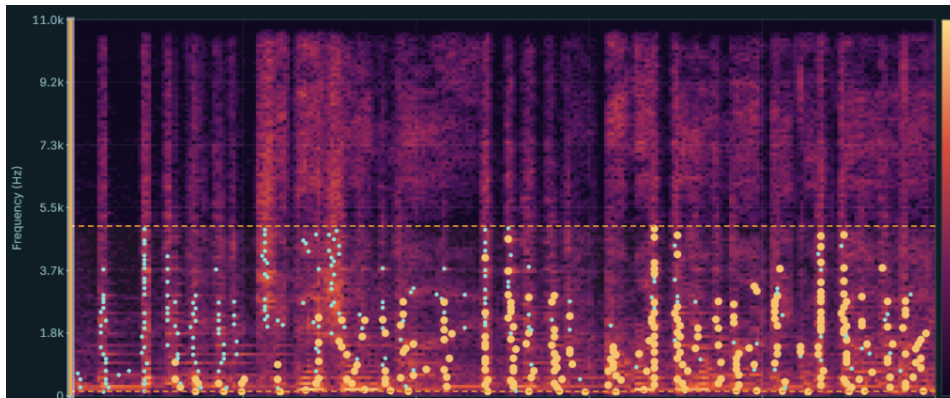


2,048-sample windows (93 ms); hop 512 (23 ms). Each window gives one column.

# Keep strong local peaks

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Each dot marks one frequency at one moment.



15 frequency bins  $\times$  9 time frames · 100–5,000 Hz;  $\geq -60$  dB · at most 60 peaks/second

# How Apollo handles edited audio

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|                        | Frequency / pitch | Time between peaks |
|------------------------|-------------------|--------------------|
| <b>Speed edit</b>      | changes           | changes            |
| <b>Pitch-only edit</b> | changes           | stays the same     |

**78 quick guesses: 39 speed + 39 pitch-only factors**

Rescale detected peaks → make new fingerprints → test the best guesses.

This fallback runs when ordinary fingerprint matching misses an edit.